

Stoquify POS 9+ Evaluation and Roadmap

Evidence-based multidisciplinary assessment • 20 July 2026

Date: 2026-07-20

Evaluation mode: read-only product evaluation and planning. No product code was changed.

1. Executive verdict

Stoquify POS is a credible, finance-aware online workstation with a stronger transaction kernel than its current cashier experience communicates. The normal sale path atomically claims the draft and active cashier session, validates tender arithmetic, issues stock through the inventory kernel, updates cash/session state, creates payment rows, posts balanced accounting entries, produces fiscal evidence, and records audit/business events. Shift close, signed public receipt access, receipt-token revocation, and tenant-derived server actions are also substantial strengths.

The complete advertised workflow is nevertheless NO-GO for 9+ and NO-GO for production offline selling. Overall score: 5.8/10, Medium-High confidence. The score is capped by unresolved P0 controls rather than averaged upward by visual polish or passing unit tests.

Strongest current capabilities

- Code-confirmed: one canonical route and service/action/hook/UI boundary: ``app/[locale]/(dashboard)/dashboard/pos``, ``components/pos``, ``hooks/posHooks``, ``actions/pos``, and ``services/pos``.
- Code-confirmed: atomic normal-sale stock, payment, drawer, accounting, fiscal, audit, and outbox effects in ``services/pos/pos.service.ts:1687-2248``.
- Code-confirmed: cash change and split-tender arithmetic, on-account credit checks, external-reference normalization, optimistic stock claims, and balanced-journal assertions.
- Code-confirmed: hardened, idempotent, evidence-rich shift close in ``services/pos/pos.service.ts:1086-1334``.
- Code-confirmed: tenant-scoped signed public receipt tokens, revocation, and public-data redaction in ``services/pos/receipt.service.ts`` and ``actions/pos/receipt-token.actions.ts``.
- Test-confirmed this run: repository typecheck passed; 11 focused POS suites passed 67/67 tests; all seven POS action suites passed 43/43 tests; the offline specialist run passed 29/29 tests across four suites.

P0 release constraints

1. ``STORE_CREDIT`` can complete a sale without an authoritative customer/store-credit balance or atomic redemption.
2. Shift opening is not concurrency-safe and lacks a database-enforced one-active-session-per-terminal invariant.
3. Electronic tenders are immediately marked paid from operator-entered references; no provider acknowledgement proves capture.
4. Electronic refunds/voids have no provider reversal and cannot be honestly represented as completed external money movement.
5. The cashier charge flow has no offline capture fallback even though the UI reports an armed degraded path.
6. Offline transport errors can strand local events in ``SYNCING``, and accepted local evidence is pruned before replay reaches a terminal accounting outcome.
7. Offline device signatures are persisted but not cryptographically verified.
8. Offline conflicts can block close but have no supported acknowledge/resolve/retry workflow.

2. Scope and methodology

Target

- Route: ``/[locale]/dashboard/pos``
- Entry: ``app/[locale]/(dashboard)/dashboard/pos/page.tsx``

- Main UI: `components/pos/ProfessionalPOSSystem.tsx`
- Adjacent surfaces: terminal management, cash drawer, cash-payment history, public receipts, inventory, reconciliation, accounting, and close assurance.

Personas

Cashier, shift supervisor, returns/refund/void approver, branch manager, accountant, owner, and support/operator.

Method

The coordinating review applied:

- `stoquify-workflow-uiux-evaluator`
- `aqstoqflow-uiux-00-orchestrator`
- `007-aqstoqflow-pos-ledger-controls`
- `014-aqstoqflow-offline-pos-sync`

Three bounded read-only specialist agents reviewed: (1) cashier UX/frontend/accessibility/market fit, (2) transaction truth/security/accounting, and (3) offline reliability/tests/release readiness. The coordinator reconciled current source, graphs, historical reports, focused test execution, typecheck, current official competitor documentation, and a local authenticated-browser attempt.

Evidence labels: Code-confirmed, Test-confirmed, Browser-observed, Report-derived, Externally sourced, Inference, and Untested gap.

3. Agent assignments and synthesis

Review lens	Result
POS UX/frontend	Desktop/tablet workstation is credible; device/network truth, offline continuity, robust query states, accessibility certification, and performance measurement are missing.
Business truth/security	Normal sale/close kernel is strong; store credit, shift-open race, provider capture/reversal, correction scope, module gates, and stubs block release claims.
Offline/tests	Server replay spine is serious; cashier offline selling is unwired and queue/outcome/signature/conflict lifecycle is unsafe. Offline score: 3.0/10.

Conflicts were resolved against current source. Historical reports and graph artifacts were treated as leads, not implementation truth.

4. Evidence reviewed and verification

Current source and required evidence

The review traced the route, `ProfessionalPOSSystem`, POS hooks/actions/services, offline queue and sync service, receipt services, RBAC mappings, Prisma models, accounting posting rules, inventory stock-event boundaries, messages, global styles, and related tests.

Required reports included the UI/UX honest review, revamp roadmap, constitution, review checklist, route maturity matrix, offline POS roadmap, July offline functionality report, POS implementation reports, sale-truth and offline-replay assurance reports, and current `what-next/` release evidence.

Graph evidence included `graphify-out/GRAPH\_REPORT.md`, `graphify-out/graph.json`, and `graphify-out/ordered-code-graph.json`. The ordered graph is dated 2026-07-14 and is structurally useful but stale relative to the current dirty worktree; source wins where they disagree.

Commands executed

Command	Result	Interpretation
`npm run typecheck`	Passed, exit 0, 167.4s	Current TypeScript workspace compiles without emit.
Focused route/component/hook/action/service/receipt tests	11/11 suites, 67/67 tests passed	Substantial online POS, shift-close, receipt, and UI contract coverage.
`npm test -- --runInBand actions/pos/___tests___`	7/7 suites, 43/43 tests passed	Current action-layer RBAC/result wiring covered by those suites.
Offline queue/sync/action/static-gate suite	4/4 suites, 29/29 tests passed	Tested offline kernel behavior, not cashier offline readiness.

Passing tests do not cover unbacked store credit, concurrent shift opening, provider acknowledgement/reversal, real-database replay races, offline cashier fallback, stale `SYNCING` recovery, per-event outcome mapping, signature verification, physical hardware, or authenticated EN/FR browser completion.

5. Current architecture and trust boundaries

```

``mermaid
flowchart LR
  UI["POS route and ProfessionalPOSSystem"] --> H["TanStack hooks"]
  H --> A["Protected server actions"]
  A --> S["POS services"]
  S --> DB["Tenant-scoped Prisma transaction"]
  DB --> INV["Inventory stock-event kernel"]
  DB --> PAY["Payments and drawer/session totals"]
  DB --> LEDGER["Accounting postings"]
  DB --> FISCAL["Fiscal and receipt evidence"]
  DB --> AUDIT["Audit, business event, outbox"]
  UI --> STATUS["Offline status strip"]
  Q["Local offline queue"] --> INGEST["Protected batch ingestion"]
  INGEST --> PENDING["Server PENDING_REPLAY"]
  PENDING --> REPLAY["Separate replay action"]
  REPLAY --> S
  ...

```

Trust-boundary assessment

- Code-confirmed: the route checks `OPERATE\_POS`; write actions derive organization/user from authenticated context.
- Code-confirmed: refund/void require dedicated permissions and authentication freshness no older than five minutes.
- Code-confirmed: normal sale commit reasserts tenant, location, terminal, session, creator, and draft-state ownership.
- Code-confirmed: module-entitlement enforcement is inconsistent: receipt-token actions enforce it, tender writes omit it, and some catalog/session gates remain observe-only.
- Inference: cash-drawer ownership is currently inferred through tenant-owned terminal/location/session relations; explicit tenant ownership would improve defense in depth.

6. Persona and job-to-be-done matrix

Persona	Primary job	Current support	Principal gap
Cashier	Open register, find items, take payment, give receipt	Strong online happy path with hotkeys, split tender, change, and proof	Misleading device/network state, no scanner pipeline, no offline charge fallback, dense secondary administration
Supervisor	Resolve exceptions and authorize corrections	Dedicated refund/void services with fresh auth	No correction UI; original active-session restriction prevents ordinary post-shift returns
Manager	Control sessions, drawers, replay, and exceptions	Terminal/drawer dashboards and evidence-rich close	Shift-open race; no conflict resolver; manager override is a placeholder
Accountant	Trust stock, payment, ledger, receipt, and close evidence	Strong normal-sale/posting/fiscal evidence	Store-credit liability unsupported; provider capture/reversal unverified; replay may outlive local evidence
Owner	See branch operational truth	Session totals, drawer state, business events	No integrated settlement/replay/exception command view
Support	Diagnose sync/device/provider failures	Server conflict and blocker evidence exists	No supported resolution workbench or support bundle

7. End-to-end workflow assessment

Journey	Current state and boundaries	Recovery/evidence	Verdict
Enter POS	Route checks 'OPERATE_POS', component fetches locations/terminals/session	Permission denial stops before render	Strong server gate; module gating inconsistent
Select location/terminal	Defaults first valid location/terminal	Query failures often collapse into empty/default UI	Refine robust states and switching safety
Open shift	Precheck then transaction creates session/drawer state	Action notifications	P0: concurrent opens can both pass precheck
Find/scan item	Search, categories, favorites/recent, stock badges	Ctrl/Cmd+K focuses search	Un-debounced; no scan terminator/auto-add
Edit cart	Add/update/remove with stock-aware clamping	Rollback/refetch and notifications	Strong basics; no undo, controlled discount, park/resume
Attach customer	Walk-in/customer search, credit warning	Client-side filtering	Add paging, masking, quick create, privacy controls
Cash/split tender	Change, exact/quick cash, multiple tenders	UI blockers and server arithmetic	Strong arithmetic; method-policy gaps
Card/mobile money/bank	Reference required and duplicate-checked	Immediately marked paid/posted	P0: evidence entry is not provider-confirmed capture
Store credit	Selectable tender and accounting mapping	No backing balance/redemption	P0: disable until authoritative instrument exists
Commit sale	Online mutation calls atomic service	Pending lock, error notification, post-sale proof	Strong online kernel; no offline fallback
Receipt	Signed public receipt/token controls; selectable channels	Delivery adapters return 'PENDING' stubs	Hide/label unavailable channels
Network loss before charge	Online mutation fails	Manual retry only	P0: no provisional capture
Offline ingest	Validates scope/hash/sequence and stores 'PENDING_REPLAY'	Certificate/conflict evidence	Good foundation; client mapping unsafe
Offline replay	Separate protected action reuses normal finalizer	Replay/blocker evidence transactional	No production trigger/worker/operator flow
Duplicate replay	Status checks and completed-sale lookup	Unit-covered	Require real Postgres race proof
Conflict	Server records conflict and close blocker	Read-only visibility	P0: no acknowledge/resolve/retry
Close shift	Serializable, idempotent, variance/evidence protected	Unit and Postgres concurrency coverage	Strong; must include provider/offline blockers

Journey	Current state and boundaries	Recovery/evidence	Verdict
Refund/void/return	Atomic full-sale services exist	Fresh auth and accounting evidence	No UI, partial return, cross-shift flow, provider reversal

8. Browser, accessibility, and hardware limitations

An authenticated browser sweep was attempted on 2026-07-20 against an isolated local Next.js server. The POS route compiled and returned HTTP 200 before session resolution redirected to login. Better Auth could not validate the stored tenant session because Prisma rejected the configured datasource URL with `P6001` in that runtime.

Consequently:

- No authenticated EN/FR POS screenshot is claimed.
- No axe result, screen-reader result, complete keyboard-only checkout, or measured authenticated overflow is claimed.
- No business state was mutated.
- The isolated server, port, build directory, and Next-added `tsconfig` include entry were removed.

Physical barcode scanner, printer, drawer, payment terminal, mobile-money provider, network-loss, browser-restart, and field-device evidence are unavailable. Accessibility, responsiveness, performance, and hardware scores therefore remain below High confidence.

9. Current competitive benchmark

Official documentation was reviewed on 2026-07-20. The relevant standard is operational transparency and recoverability, not imitation.

Product	Documented principle	Transferable requirement
[Square](https://squareup.com/help/us/en/article/7777-process-card-payments-with-offline-mode)	Configured offline modes, supported/unsupported matrix, disruption detection, countdown, pending history, upload window, merchant risk	Show real state, limits, pending age, risk, and recovery outcome
[Shopify POS](https://help.shopify.com/en/manual/sell-in-person/shopify-pos/selling-offline/offline-payments)	Staff permission, compatible readers, limits, explicit offline checkout, pending/paid/declined badges	Make eligibility, device support, authority, and terminal outcomes explicit
[Lightspeed Retail](https://x-series-support.lightspeedhq.com/hc/en-us/articles/25534272395163-Selling-in-offline-mode)	Constrained offline surfaces and unsynced-sale retry visibility	Provide an operator workbench, not only an aggregate strip
[Toast](https://support.toasttab.com/en/article/Using-Toast-in-Offline-Mode)	Cause-aware offline banner and documented local-device limitations	State exactly what continues during each outage
[Odoo POS](https://www.odoo.com/app/point-of-sale-features)	Device portability, peripherals, split payments, accounting integration	Certify a concrete device matrix while retaining ledger integration
[Clover](https://docs.clover.com/dev/docs/handling-offline-payments)	Configurable offline risk controls and queued handling	Add tender policy, limits, and explicit liability
[Yoco](https://www.yoco.com/za/point-of-sale/handheld-pos/)	African-market handheld/counter hardware, connectivity, payments, stock, reporting	Validate field hardware, connectivity, and local ergonomics

10. Dimension-by-dimension scorecard

#: 1
Dimension: Cashier speed and efficiency
Current: 6.5
Confidence: Medium
Main gap: No scan-to-add proof; dense secondary content; broad refetches
Objective 9+ condition: Measured scan-to-cart and cart-to-paid budgets pass under peak load
#: 2
Dimension: Visual hierarchy and density

Current: 7.0 Confidence: Medium Main gap: Setup, proof, receipt history, and tender compete on one 2,305-line screen Objective 9+ condition: Checkout remains primary; secondary proof/admin moves to drawers/workbenches
#: 3 Dimension: Catalog search and scanning Current: 6.0 Confidence: Medium Main gap: Un-debounced search; no scanner buffer/terminator Objective 9+ condition: Certified scanner flow and large-catalog latency budget
#: 4 Dimension: Cart accuracy/editability Current: 7.0 Confidence: Medium-High Main gap: No controlled discount/override, park/resume, undo, notes Objective 9+ condition: Complete permissioned edit/exception model with tests
#: 5 Dimension: Customer usability/data safety Current: 6.5 Confidence: Medium Main gap: Whole dataset fetch; unmasked contact/revenue exposure Objective 9+ condition: Server paging, minimization/masking, purpose-based reveal
#: 6 Dimension: Tender, split, cash, change Current: 5.5 Confidence: High Main gap: Unbacked store credit; unverified electronic capture Objective 9+ condition: Authoritative tender policy and full matrix tests
#: 7 Dimension: Shift and drawer control Current: 6.5 Confidence: High Main gap: Concurrent shift opening can race Objective 9+ condition: Atomic station claim and DB-enforced one-active-session invariant
#: 8 Dimension: Receipt and fiscal evidence Current: 6.0 Confidence: High Main gap: Delivery stubs; offline receipt absent Objective 9+ condition: Real adapters and truthful provisional/final delivery state
#: 9 Dimension: Refund/void/return safety Current: 4.0 Confidence: High Main gap: No UI, partial/cross-shift flow, or provider reversal Objective 9+ condition: Permissioned post-shift/partial correction with tender disposition

#: 10 Dimension: Offline resilience/recovery Current: 3.0 Confidence: High Main gap: No cashier fallback; unsafe queue lifecycle/signatures/conflicts Objective 9+ condition: Field-proven capture through exact-once replay and final receipt
#: 11 Dimension: Robust and conflict states Current: 5.0 Confidence: High Main gap: Query failures collapse; `SYNCING` can strand; no resolver Objective 9+ condition: Every state has truthful recovery, owner, SLA, terminal outcome
#: 12 Dimension: Keyboard/touch/scanner Current: 5.5 Confidence: Medium Main gap: Partial hotkeys/touch sizing; fabricated readiness Objective 9+ condition: Keyboard-only checkout and certified device matrix
#: 13 Dimension: Accessibility Current: 5.5 Confidence: Low-Medium Main gap: Missing accessible names/announcements and current audit Objective 9+ condition: Zero critical/serious axe issues plus keyboard/screen-reader completion
#: 14 Dimension: Responsiveness Current: 6.0 Confidence: Low-Medium Main gap: Mobile stacking/density unverified Objective 9+ condition: Certified target viewports with no overflow or unsafe reordering
#: 15 Dimension: Performance posture Current: 5.0 Confidence: Medium Main gap: Monolith, un-debounced queries, broad invalidation, no budget Objective 9+ condition: p95 search/add/charge and large-catalog budgets pass
#: 16 Dimension: Localization/currency/tax/rounding Current: 6.5 Confidence: Medium Main gap: Unaccented French; USD fallback; `0.01` assumptions for XAF Objective 9+ condition: Reviewed EN/FR and country-pack-derived minor-unit/cash rounding
#: 17 Dimension: Security/RBAC/tenant isolation Current: 6.5 Confidence: High Main gap: Inconsistent module gates; no offline authenticity; capability-blind UI

Objective 9+ condition: Uniform entitlement/RBAC/fresh-auth and cryptographic device identity
#: 18 Dimension: Inventory/accounting truth Current: 6.5 Confidence: High Main gap: Strong normal path; store-credit/provider/offline truth incomplete Objective 9+ condition: All tender/correction/offline paths reconcile and post atomically
#: 19 Dimension: Manager observability/audit Current: 5.0 Confidence: High Main gap: No correction/replay/provider exception workbench Objective 9+ condition: Manager can explain, resolve, approve, evidence every blocker
#: 20 Dimension: Test and release readiness Current: 5.0 Confidence: High Main gap: Many tests pass; key real-DB/browser/device/provider journeys absent Objective 9+ condition: Full gate matrix passes in intended environment
#: 21 Dimension: OHADA/target-market fit Current: 6.5 Confidence: Medium Main gap: French, XAF, mobile money, offline, provider claims unverified Objective 9+ condition: Country-pack and bilingual field acceptance passes
#: 22 Dimension: Competitive differentiation Current: 6.0 Confidence: Medium-Low Main gap: Accounting proof leads; continuity/hardware/returns lag Objective 9+ condition: Finance moat plus competitor-grade continuity and device truth

Unweighted mean is approximately 5.8. Critical P0 controls cap the verdict regardless of arithmetic averaging.

11. UX and visual-system findings

What works

- Code-confirmed: coherent authenticated dark dashboard tokens and a desktop catalog/checkout split at components/pos/ProfessionalPOSSystem.tsx:1051-1122.
- Code-confirmed: location/terminal setup, shift state, product discovery, cart, tender, last-sale proof, and offline status exist in one canonical surface.
- Code-confirmed: Ctrl/Cmd+K, F4, F8, and Escape support fast focus movement at ProfessionalPOSSystem.tsx:540-563.
- Code-confirmed: exact-cash/quick-cash, grid/list, favorites/recent, touch toggle, stock visibility, split tender, and receipt selection are present.

Main friction

- Code-confirmed: ProfessionalPOSSystem.tsx is approximately 2,305 lines and owns query orchestration, transaction interaction, receipt administration, local favorites/recent state, shortcuts, dialogs, and full page rendering.
- Code-confirmed: receipt-token history occupies cashier checkout space even though it is a permissioned administrative task.
- Code-confirmed: park sale and manager override are clickable backend-gated placeholders at ProfessionalPOSSystem.tsx:1309-1316.
- Code-confirmed: catalog search changes the query on every keystroke without a debounce; customer data is fetched broadly then filtered client-side.
- Code-confirmed: favorites/recent exist only in component memory and reset with the session/page.
- Inference: high information density supports trained operators but burdens first-run cashiers and narrow screens because setup, proof, and exception administration compete with the selling loop.

12. Accessibility, responsive, keyboard, touch, and hardware findings

- Code-confirmed: visible labels, dialog primitives, aria-invalid, a variance alert, focus restoration, numeric input modes, and several keyboard accelerators are positive foundations.
- Code-confirmed: the cart delete action is icon-only without an explicit accessible name at ProfessionalPOSSystem.tsx:1691-1700; other icon actions rely on title.
- Code-confirmed: catalog skeletons do not expose aria-busy/status announcements, and many query failures have no inline recovery.
- Code-confirmed: touch mode enlarges selected controls, not a governed end-to-end touch target system.
- Code-confirmed: network status is always rendered online and scanner readiness is inferred from canSell at ProfessionalPOSSystem.tsx:609-614; the header separately announces scanner readiness without detection at 1075-1079.
- Untested gap: no authenticated axe, screen-reader, contrast, focus-order, physical scanner, printer, drawer, payment terminal, or measured mobile/tablet test completed in this run.

The product must either certify supported phone/tablet/device modes or state unsupported configurations explicitly.

13. Business-truth and accounting-control findings

Strong normal-sale kernel

- Code-confirmed: tender arithmetic rejects short payment and non-cash overpayment while calculating cash change at services/pos/pos.service.ts:355-399.
- Code-confirmed: commit uses a transaction and compare-and-set claims on the draft/session at services/pos/pos.service.ts:1687-1825.
- Code-confirmed: on-account sales require a customer and enforce credit limits at 1827-1851.
- Code-confirmed: tracked stock issues use the inventory stock-event kernel at 1854-1885.
- Code-confirmed: cash updates shift/drawer truth and writes drawer transactions at 1888-1975.
- Code-confirmed: sale/payment postings and balanced-journal assertions occur inside the transaction at 2031-2080.
- Code-confirmed: fiscal document, audit, and business-event/outbox evidence are created at 2082-2175.

Financial release blockers

- P0 - store credit: the UI and schema accept STORE_CREDIT, and commit creates a paid payment/liability posting, but there is no authoritative instrument, balance lookup, redemption, decrement, or reversal.
- P0 - electronic capture semantics: card/mobile-money/bank references are normalized and duplicate-checked, but no provider is contacted. These are externally confirmed/manual evidence, not verified capture.
- P0 - provider reversal: refund and void update internal rows, stock, cash, and accounting but do not call a card/mobile-money/bank reversal adapter.

- P0/P1 - correction scope: full-sale refund/void requires the original session to remain active and owned by the current cashier; ordinary post-shift supervisor returns are blocked.
- P1 - receipt delivery: print/email/SMS/WhatsApp adapters are stubs returning PENDING; the public signed receipt itself is real.
- P1 - cash refunds: add an explicit cash-availability/drawer-authority check for large cash refunds.

14. Security, RBAC, privacy, and tenant-isolation findings

Strengths

- Tenant and actor identity come from protected action context.
- Sale/session/location/terminal queries reapply organization and user boundaries.
- Refund/void require dedicated permissions and fresh authentication.
- Receipt-token search/list/revoke is tenant-scoped; public receipts redact customer contact data.
- Offline server actions derive org/user and assert active device/terminal/location scope.

Gaps

- P0: sale/session/catalog module-entitlement enforcement is missing or observe-only while receipt-token actions enforce it.
- P1: the shared UI renders shift/correction-like controls without a proven capability model; server denial remains safe but creates avoidable operator failure.
- P0 offline: signature fields exist but ingestion does not verify a device signature/public key.
- P1 privacy: cashier customer search exposes phone/email/lifetime revenue without a demonstrated field-minimization or masking policy.
- P2 defense in depth: cash drawer has no explicit organizationId; ownership is indirect through terminal/location/session.

15. Offline reliability and recovery findings

Backend replay controls are much stronger than the cashier journey, but the current workflow is not usable offline.

1. No cashier fallback: ProfessionalPOSSystem.tsx:870-909 calls only online commitSale.mutateAsync; no production consumer wires useEnqueueOfflinePOSEvent into charge.
2. Stranded SYNCING: useOfflineSync.ts:214-223 marks entries before awaiting transport; a thrown transport error bypasses failure update. The selector retries only QUEUED/FAILED.
3. Premature local pruning: successful ingest stores server events as PENDING_REPLAY, yet the hook marks all local events SYNCED and prunes when aggregate conflicts are zero without invoking replay.
4. No per-event outcome: mixed batches receive aggregate counts, so accepted and conflicted local entries cannot be mapped safely.
5. No signature verification: payload/hash-chain checks exist, but persisted signatures are not cryptographically verified.
6. Fragile durability: plain localStorage, silent reset on malformed content, read-modify-write without multi-tab locking, and no quota/corruption recovery.
7. No conflict resolution: models and read views support conflict state, but no production acknowledge/resolve/retry operation exists.
8. Misleading readiness: the strip announces the degraded path as armed without charge fallback, replay trigger, readiness snapshots, signature validation, or conflict resolution.

Offline strengths

- Stable JSON hashing and SHA-256 hash-chain continuity.
- Device sequence, idempotency, previous-hash, payload-hash, and scope checks.
- Provisional-only fiscal guard.

- Replay reuses commitPOSSale, blocks failures with evidence, and recovers already-completed sales.
- Server replay/blocker evidence is transactional.
- Close assurance can detect pending/conflicted replay evidence.

16. State and failure-mode matrix

State/event	Current behavior	Required recovery
Loading catalog	Skeleton cards	Add aria-busy, preserved last-safe data, timeout/retry
Empty/no shift	Guided empty state	Keep; clarify setup owner and permission
Query failure	Often empty/default UI	Inline cause, retry, stale indicator, escalation
Permission denied	Server rejects	Capability-aware UI plus intentional denied state
Sale pending	Charge disabled and notification shown	Add idempotency/retry outcome reference
Sale business-rule failure	Error notification	Preserve cart/tender and guide correction
Network failure	Online mutation error	Classify eligible outage vs business denial; provisional capture only when safe
Local QUEUED	Eligible for flush	Show age, attempts, device/session/tender risk
Local SYNCING	Not retry-eligible if stranded	Lease/timeout recovery to FAILED
Local SYNCED	Pruned after ingest	Retain until replay terminal state/reconciliation
Local CONFLICT	Retained	Supported acknowledge/resolve/retry with audit
Server PENDING_REPLAY	Close blocker	Worker/operator trigger, SLA, ownership
Server replayed/duplicate	Terminal economic state	Link sale, receipt, posting, document hash
Blocked/quarantined	Evidence retained	Actionable reason, remediation, approval, retry/terminal close
Receipt delivery pending	Stub-provider response	Truthful unavailable/pending state and provider retry
Shift variance	Explanation required	Keep; add threshold policy/approval where needed

17. Test coverage and evidence-gap matrix

Area	Current evidence	Missing release evidence
Route permission	Focused test passed	Authenticated browser allow/deny with module entitlement
Cart/tender/commit	Service/UI unit coverage passed	Full split-tender matrix, duplicate click, provider failure, large catalog
Shift close	Unit and existing Postgres concurrency tests	Intended deployment DB execution in this run
Shift open	Basic action/service coverage	Concurrent open and duplicate drawer Postgres tests
Receipt tokens	Component/action/service coverage passed	Authenticated receipt lifecycle and real delivery adapters
Refund/void	Atomic service/accounting tests exist	UI, cross-shift manager, partial/line return, provider reversal
Offline queue/sync	29 focused tests passed	Hook rejection, stale SYNCING, corruption, quota, multi-tab, mixed ACK
Replay	Service mock coverage	Real Postgres exact-once concurrent replay and rollback
Signature	Schema/status vocabulary	Signing, invalid signature, rotation/revocation
Accessibility/responsive	Source inspection only	Authenticated axe, keyboard, screen reader, target viewports

Area	Current evidence	Missing release evidence
Hardware/provider	Labels/stubs only	Certified scanner/printer/drawer/terminal/mobile-money field runs
Performance	Memoization/query caching	p95 interaction, catalog scale, low-network, offline volume budgets

The current static offline readiness artifact reports ready, but its gate is primarily source-string inspection and does not prove these runtime behaviors.

18. Keep / Refine / Split / Move / Replace / Remove

Decision	Target	Rationale
Keep	Atomic commitPOSSale, compare-and-set claims, inventory kernel, posting rules, fiscal/audit/outbox evidence	Strongest product/control foundation
Keep	Hardened shift close and signed receipt registry	Evidence-rich, idempotent patterns worth reusing
Refine	Tender policy, robust states, permissions, localization, invalidation, touch targets	Foundations exist; controls/ergonomics incomplete
Refine	Shift open	Add serialized station claim and one-active invariant
Split	ProfessionalPOSSystem	Separate register shell, catalog, cart/tender, proof, and admin without moving truth client-side
Move	Receipt-token history	Permissioned receipt/history workspace or secondary drawer
Move	Returns/refunds/voids	Supervisor correction workbench with fresh auth and original-tender disposition
Replace	StubReceiptDeliveryProvider	Cannot support delivery claims
Replace	localStorage queue	IndexedDB transactional store with leases, corruption handling, recovery
Remove/disable	Store credit until backed	Current option can create unsupported liability
Relabel	Scanner/network/offline readiness	Never claim readiness without objective detection

19. Ideal future POS page anatomy

1. Register status bar: tenant/location/terminal/cashier/shift, actual network/device state, offline eligibility, pending replay age.
2. Primary selling workspace: scanner/search, categories, product grid/list, stock truth, fast cart manipulation.
3. Checkout sidecar: customer, totals, tender policy, change, receipt choice, one decisive charge action.
4. Exception layer: inline validation, stale/conflict status, permission-safe supervisor escalation.
5. Post-sale proof drawer: sale number, tender disposition, stock event, journal entries, fiscal/receipt status.
6. Separate supervisor workbenches: returns/corrections; receipt history/token governance; offline replay/conflicts; X/Z and close readiness.
7. Shared robust states: loading, empty, denied, locked, stale, partial, offline, retrying, replayed, blocked, and terminal outcomes.

20. Prioritized P0-P3 implementation roadmap

P0 — release blockers

Initiative: Enforce tender truth

Problem and evidence: STORE_CREDIT is selectable but no balance or liability subledger is enforced; electronic tenders can become PAID from operator-entered references.

User/business impact: False settlement, misstated liabilities, fraud exposure.

Effort / dependency / owner: L / payment and accounting domains / Payments + Accounting.

Acceptance criteria and verification: Disable unsupported tender until a reserve ledger exists; electronic PAID requires signed provider acknowledgement or an explicit controlled pending/manual-settlement state. Unit, integration, reconciliation, and negative authorization tests pass.

Rollback boundary: Feature flags per tender; preserve cash sales.

Initiative: Serialize shift opening

Problem and evidence: Application checks do not create a database-level one-active-session invariant.

User/business impact: Concurrent drawers and ambiguous accountability.

Effort / dependency / owner: M / schema migration / POS backend.

Acceptance criteria and verification: Concurrent open attempts yield exactly one active session; losing request returns a recoverable conflict; migration and race test prove invariant.

Rollback boundary: Migration must be reversible; retain prior sessions unchanged.

Initiative: Implement provider reversal truth

Problem and evidence: Refund/void paths do not execute or prove processor reversals.

User/business impact: Customer can be promised money that was not returned.

Effort / dependency / owner: L / provider adapters and reconciliation / Payments.

Acceptance criteria and verification: Refund state separates requested, accepted, settled, failed; provider reference and evidence are immutable; void/refund cannot finalize without eligible reversal or approved exception.

Rollback boundary: Adapter flag and manual exception workflow; never erase evidence.

Initiative: Make offline selling real and bounded

Problem and evidence: Active checkout has no offline charge fallback.

User/business impact: Connectivity loss stops selling despite offline claims.

Effort / dependency / owner: XL / tender policy and device capability / POS + Payments.

Acceptance criteria and verification: Eligibility is tender/device/country/risk aware; operator sees limits before charging; offline transaction is durable before confirmation; unsupported paths fail closed. Scenario, device-loss, and power-loss tests pass.

Rollback boundary: Global kill switch plus per-store limits.

Initiative: Repair offline queue lifecycle

Problem and evidence: Transport failure can strand SYNCING; client prunes after ingestion before terminal replay; localStorage queue lacks leases and robust corruption recovery.

User/business impact: Silent loss, duplicate attempts, unrecoverable limbo.

Effort / dependency / owner: XL / IndexedDB schema and replay API / Offline platform.

Acceptance criteria and verification: Durable per-event states; stale leases recover; acknowledgement is terminal-event-specific; items remain until applied/rejected; idempotent multi-tab and restart tests pass.

Rollback boundary: Dual-read migration and queue export; old queue remains readable during rollout.

Initiative: Verify device authenticity

Problem and evidence: Signature fields exist but service verifies hashes rather than cryptographic device signatures.

User/business impact: Forged offline events may be accepted.

Effort / dependency / owner: L / device key lifecycle / Security + Offline.

Acceptance criteria and verification: Registered device key verifies canonical payload; replay/expired/revoked keys fail; rotation and audit trails tested.

Rollback boundary: Verification enforcement shadow mode, then staged fail-closed.

Initiative: Add conflict resolution

Problem and evidence: No cashier/manager acknowledge-resolve-retry flow exists.

User/business impact: Conflicts remain operationally stuck.

Effort / dependency / owner: L / replay outcome model / POS UX + Offline.

Acceptance criteria and verification: Every conflict has reason, owner, permitted action, audit record, and terminal outcome; retry is idempotent.

Rollback boundary: Read-only diagnostics remain available if actions are disabled.

Initiative: Normalize module entitlements

Problem and evidence: POS entry and subordinate surfaces do not consistently prove the same entitlement boundary.

User/business impact: Unauthorized use and packaging leakage.

Effort / dependency / owner: M / module policy / Platform + Security.

Acceptance criteria and verification: Route, action, service, offline replay, and navigation all deny consistently; tenant isolation and negative tests pass.

Rollback boundary: Central gate flag; no destructive data changes.

P1 — operational completeness

Initiative	Outcome	Effort / owner	Acceptance criteria
Supervisor returns workbench	Partial, cross-shift, tender-aware returns with cash-availability and approval controls.	XL / POS + Accounting.	Original quantities and prior refunds cap eligibility; cash shortage and provider failure are explicit; audit and ledger tie out.
Device adapter layer	Real scanner, printer, cash-drawer, and terminal state instead of hard-coded indicators.	L / POS platform.	Capability detection, connect/reconnect, degraded modes, and test harness cover supported devices.
Resilient cashier shell	Clear loading, empty, denial, retry, and failure states with robust query/error handling.	M / Frontend.	No ambiguous blank states; failed mutations preserve cart; retry is safe; keyboard and screen-reader checks pass.
Decompose the monolith	Split the 2,305-line POS component by domain and move history/administration out of the selling path.	L / Frontend.	Behavioral parity tests pass; sell path bundle and render latency do not regress.
Receipt delivery truth	Use real providers or hide unavailable email/SMS choices; support retry and proof.	M / Notifications.	UI reflects provider capability; delivery outcomes and retries are observable and audited.
Locale and money correctness	Country-specific minor units, rounding, tax labels, and complete French copy.	M / Localization + Accounting.	Golden-money tests and locale review pass for each enabled country pack.
Offline operations workbench	Surface queue, replay, close dependencies, conflicts, and escalation to authorized staff.	L / POS UX.	Operators can explain every queued item and recover without database access.

P2 — scale and assurance

Initiative	Outcome	Acceptance criteria
Provider reconciliation automation	Settlement, fees, refunds, and exceptions tie to ledger evidence.	Daily reconciliation produces explained variance and close blockers.
Field certification matrix	Supported browser, device, printer, terminal, and network combinations are evidence-backed.	Each supported combination has repeatable certification artifacts.
POS observability and SLOs	Latency, decline, retry, queue age, replay, conflict, and fiscal failure are measurable.	Dashboards and alerts have owners, thresholds, and runbooks.
Explicit drawer tenancy	Drawer/device ownership is modeled rather than inferred.	Cross-terminal concurrency and handover tests pass.

P3 — differentiation

Initiative	Outcome	Guardrail
Handheld workflow	Purpose-built compact selling and line-busting experience.	Do not ship until device certification and offline boundaries are complete.
Operational analytics	Shift, product, tender, and exception insights for managers.	Analytics must derive from trusted states, not UI events.
Anomaly assistance	Explain suspicious refunds, overrides, or queue patterns.	Advisory only; human decision, evidence, and appeal remain visible.

21. Implementation sequence and dependencies

Freeze claims and unsupported tenders → shift and entitlement invariants → provider acknowledgement/reversal plus durable offline queue/device identity → offline tender policy and conflict workbench → returns workbench and field certification →

reconciliation and close assurance → 9+ release candidate.

Recommended execution order:

1. Stop overstating capability: hide unsupported store credit, mark hard-coded device states honestly, and document the current online-only sell boundary.
2. Establish database and entitlement invariants before adding UI breadth.
3. Model provider acknowledgements, reversals, and reconciliation outcomes.
4. Replace the offline queue lifecycle and implement device-key verification.
5. Wire bounded offline tender capture and conflict recovery.
6. Build supervisor returns, receipt delivery, and device adapters.
7. Complete accessibility, localization, performance, field-device, and close-assurance certification.

22. Verification plan and evidence requirements

Layer	Required proof
Static and unit	Typecheck; money, tax, tender, authorization, reducer, and state-machine tests.
Database	Concurrent shift-open test; tenant isolation; uniqueness; idempotency; migration forward/backward rehearsal.
Service integration	Sale atomicity, provider acknowledgement, reversal, stock, cash, ledger, fiscal, outbox, and reconciliation tests.
Offline chaos	Power loss, browser kill, multi-tab races, malformed storage, expired lease, duplicate replay, reordered events, revoked key, intermittent network, and server restart.
Browser UX	Authenticated Chromium/WebKit/Firefox runs at desktop and compact widths; loading/empty/error/denied/offline/conflict states; cart preservation.
Accessibility	Automated axe plus manual keyboard, focus order, visible focus, screen-reader names, contrast, zoom/reflow, and error-announcement evidence.
Hardware	Certified scanner, printer, drawer, and payment terminal matrix with disconnect/reconnect and paper/device-failure scenarios.
Financial assurance	Golden journal fixtures, refund/void settlement evidence, inventory movement tie-out, drawer close, fiscal numbering, and provider settlement reconciliation.
Operations	SLO dashboards, alert drills, incident runbooks, queue export/recovery, rollback rehearsal, and named owners.

Current passing tests are regression evidence for the existing implementation; they are not evidence that the missing P0 capabilities exist.

23. Release gates and no-go conditions

Current decision

NO-GO for a Stoquify POS 9+ claim and NO-GO for production offline selling. The online cash-ledger-stock-fiscal kernel may continue through controlled internal testing, but broad production claims must remain bounded to proven capabilities.

Mandatory gates

- G1 — Money truth: no unsupported tender; provider-confirmed payment and reversal states; reconciliation proof.
- G2 — Session truth: database-enforced single active session per declared drawer/store boundary.
- G3 — Offline durability: restart-safe durable queue, leases, per-event terminal acknowledgement, idempotent replay, and conflict resolution.
- G4 — Device trust: cryptographic signature verification, key rotation/revocation, and tenant/device authorization.
- G5 — Access truth: consistent module entitlement, fresh authorization for privileged actions, and negative tenant tests.

- G6 — Accounting truth: sale/refund/void/close journals and stock movements tie out with no unexplained suspense.
- G7 — Operator safety: authenticated browser evidence for all state classes, accessibility checks, and certified hardware recovery.
- G8 — Operability: telemetry, alert ownership, runbooks, rollback rehearsal, and support escalation are ready.

Any failed mandatory gate is a release blocker. A high aggregate score cannot compensate for a failed financial, offline, security, or tenant-isolation gate.

24. Residual risks, limitations, and open decisions

- Authenticated browser and assistive-technology evidence is missing because the isolated audit environment could not establish the required database session. Visual and accessibility observations are therefore code-evidence-based and carry lower confidence.
- The intended scope of STORE_CREDIT must be decided: remove it, integrate an external wallet, or implement a controlled liability ledger.
- The supported offline tender set, amount/time caps, country rules, device ownership, and risk acceptance require explicit product and compliance decisions.
- Define whether the one-active-session invariant belongs to tenant/store, register, drawer, device, or cashier; the database constraint must match the chosen physical-control model.
- Define refund policy for partial quantities, cross-location returns, cash availability, provider outages, fees, foreign currency, and fiscal documents.
- Choose supported printers, scanners, terminals, browsers, and compact devices before claiming compatibility.
- Current benchmark comparisons are directional: vendor capabilities vary by country, hardware, processor, configuration, and plan.

25. Recommended next prompt

Execute only the P0 hardening program from the Stoquify POS 9+ evaluation dated 2026-07-20. Begin with a read-only evidence refresh and convert each P0 item into an implementation-ready work package with exact files, schema changes, API/state contracts, security boundaries, migration and rollback plans, acceptance tests, and release evidence. Use separate agents for payment/accounting truth, offline durability/device identity, shift/session concurrency, module entitlement, and POS UX recovery. Do not implement until dependencies and ownership are reconciled into one approved sequence. Preserve unrelated working-tree changes, prohibit destructive Git operations, and treat any failed money, tenant, authorization, device-signature, replay-idempotency, or reconciliation test as a no-go.

Bottom line: Stoquify already has a credible transactional core, but it is not yet a trustworthy 9+ POS. The fastest path is to narrow claims, close the payment/session/offline trust gaps, and certify recovery behavior before adding breadth or visual polish.